

Auteur: Noran Ben Said
Studierichting: Informatica
Oefeningenreeks 7C nr. 1.3

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}(x, y) \mapsto \text{Bgtan}(xy) + \text{Bgcot}(xy)$$

$$f(x, y) = \text{Bgtan}(xy) + \text{Bgcot}(xy)$$

$$\begin{aligned}\nabla f(x, y) &= \left(\frac{y}{1 + (xy)^2} - \frac{y}{1 + (xy)^2}, \frac{x}{1 + (xy)^2} - \frac{x}{1 + (xy)^2} \right) \\ &= (0, 0)\end{aligned}$$

$$D^2 f(x, y) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

References